

Nauta® Mixing Deep Drive

by Chris Paulsworth

- The presentation will start at 2:00 PM Eastern Time (USA)
- Hosokawa panelist will not be able to hear you, however you can submit questions using the dialog box to be answered at the conclusion of the presentation. If your question is not answered, a Hosokawa representative will respond offline.
- This presentation will be recorded and available on the Hosokawa Micron Powder Systems' Website within 48 hours. Please share with colleagues and co-workers.
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HOSOKAWA MICRON POWDER SYSTEMS

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2. How Nauta® Mixers Mix
3. Component Breakdown
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The beginning

UNITED STATES PATENT OFFICE

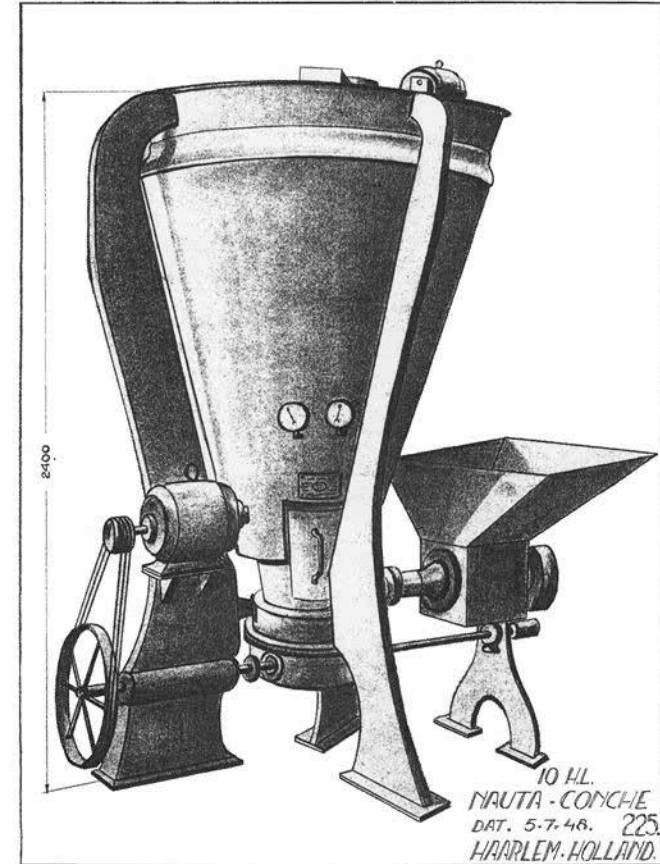
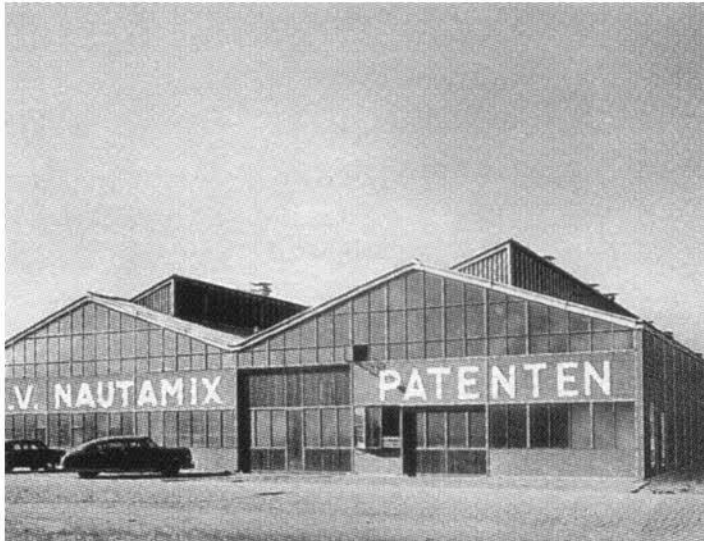
2,345,063

**MIXING APPARATUS, ESPECIALLY ADAPTED
FOR MIXING PULVERIZED MATERIALS**

Johannes Ewardus Nauta, Overveen, Netherlands;
vested in the Allen Property Custodian

Application October 31, 1940, Serial No. 363,778
In the Netherlands November 13, 1939

6 Claims. (Cl. 259—40)



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The original design

March 28, 1944.

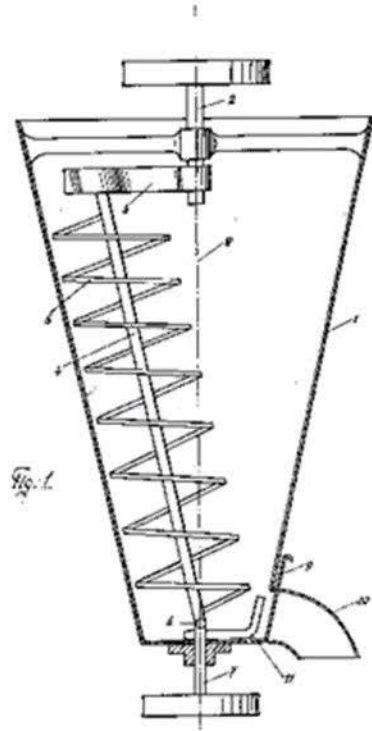
J. E. NAUTA

2,345,063

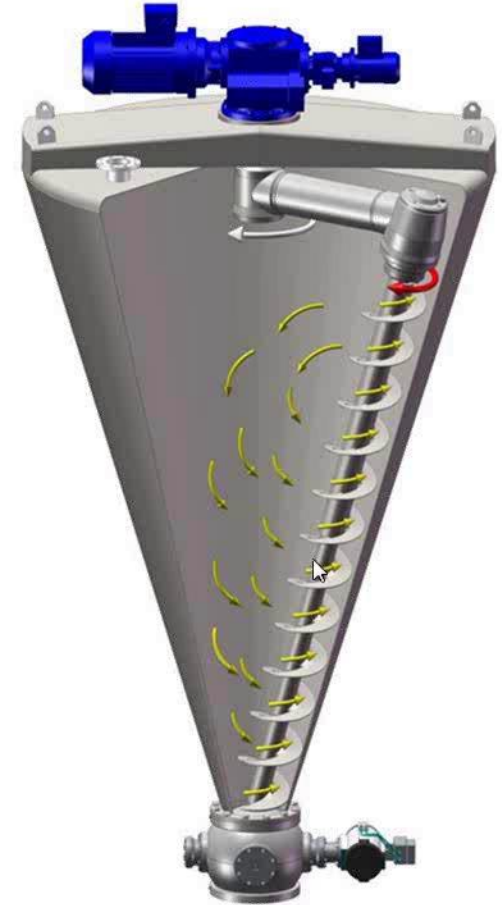
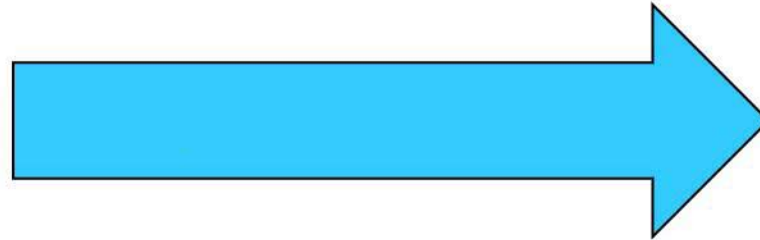
MIXING APPARATUS, ESPECIALLY ADAPTED FOR MIXING FULVERIZED MATERIALS

Filed Oct. 31, 1940

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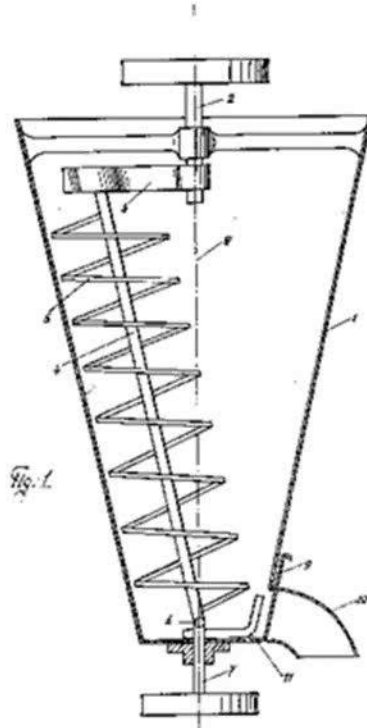
Inventor
Johannes E. Nauta
by *Alfred J. Nauta*
Att'y.



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Two deigns?

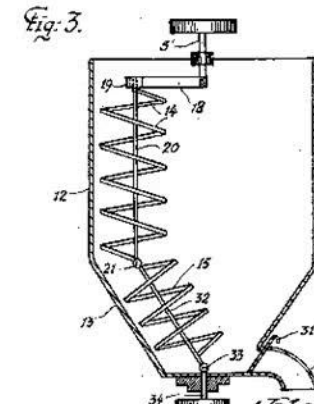
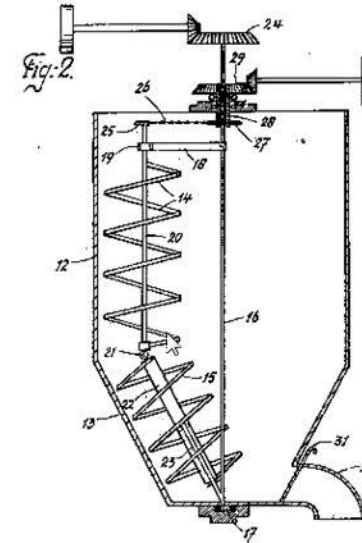
March 28, 1944. J. E. NAUTA 2,345,063
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Inventor
Johannes E. Nauta
by *[Signature]*
Atty.

&

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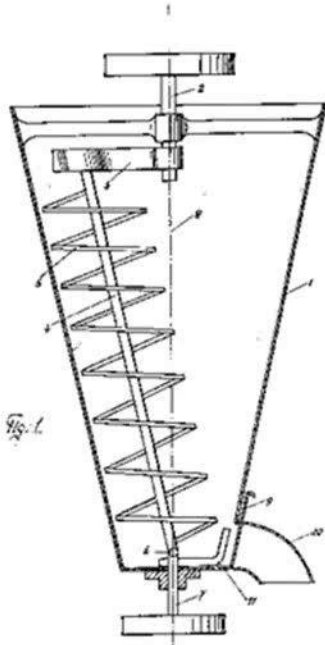
Inventor
Johannes E. Nauta
by *[Signature]*
Atty.



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73 years of continuous improvement

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Inventor
Johannes D. Nauta
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Att'y.



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Over the years



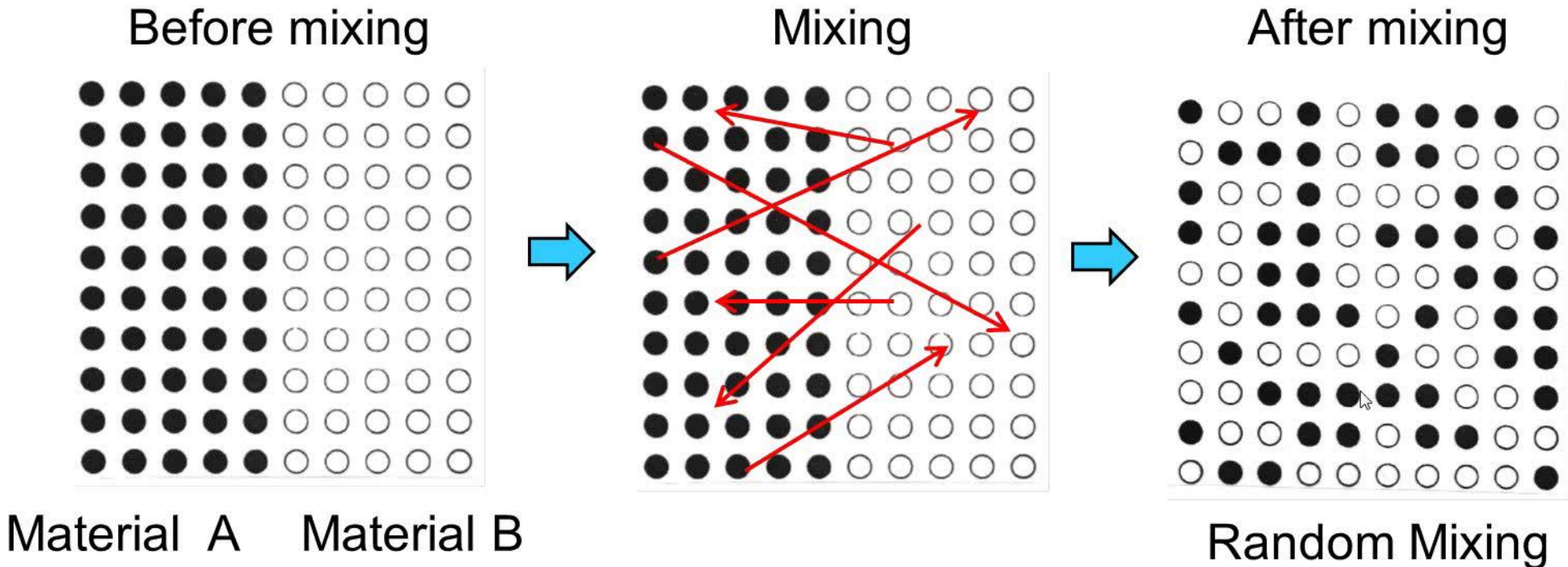
*Nautamix + Machinefabrik Vrieco + ISEM ...
+ Schugi + Hosokawa Micron
= Hosokawa Micron B.V.*



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How Mixers Work

The Mixing Process is about transporting particles

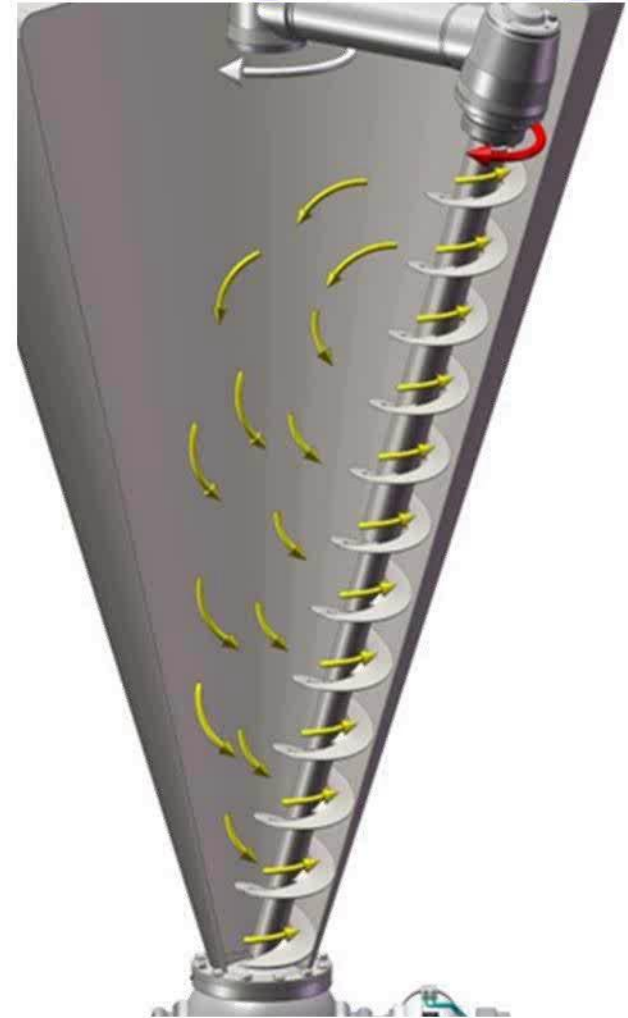
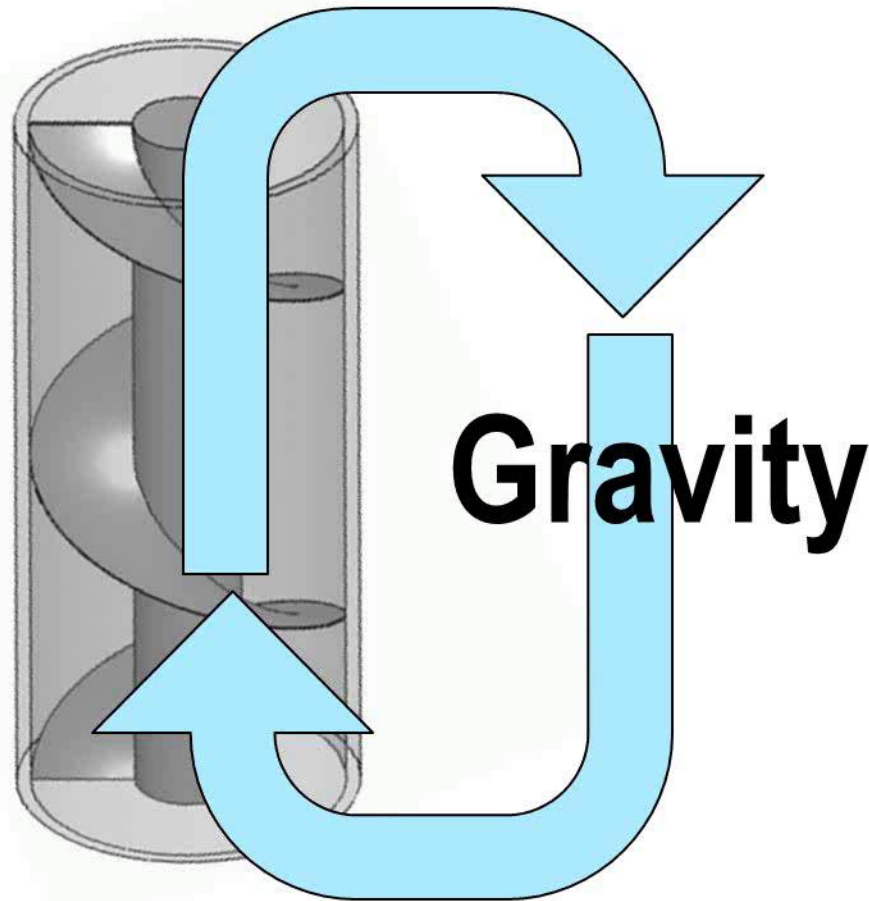


Why Nauta® Mixers?

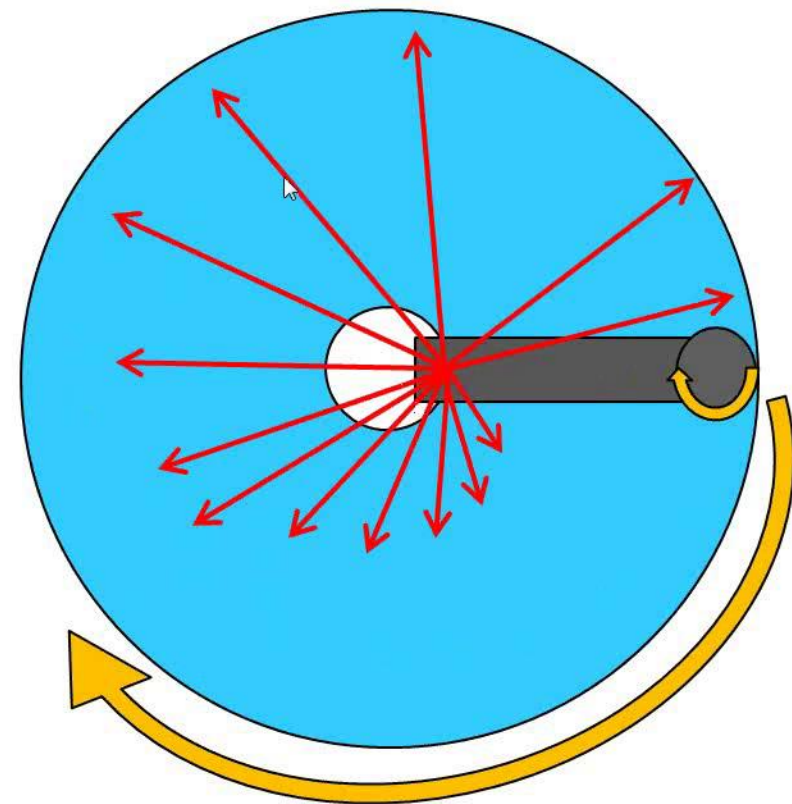
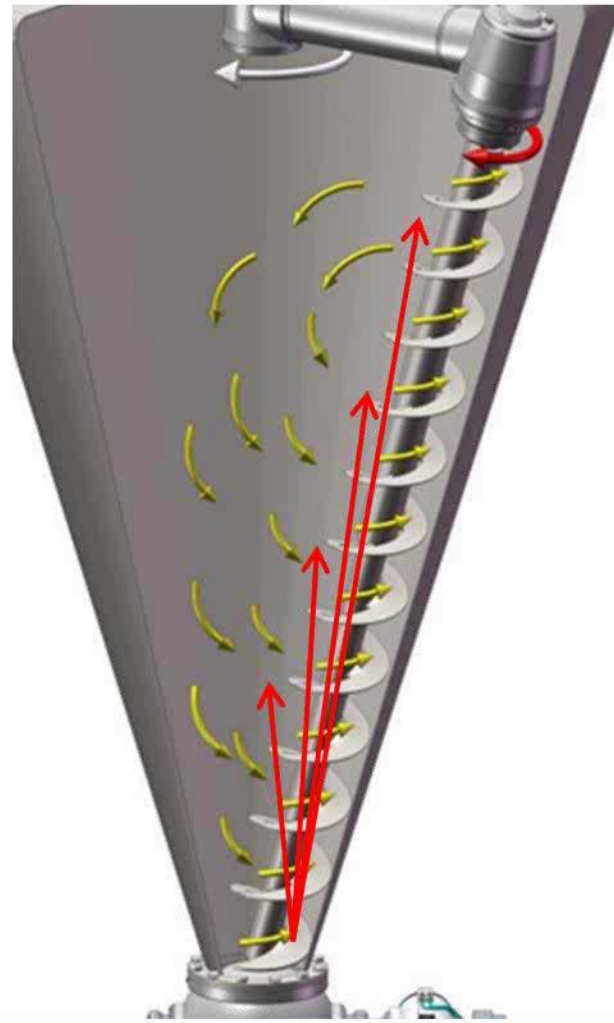
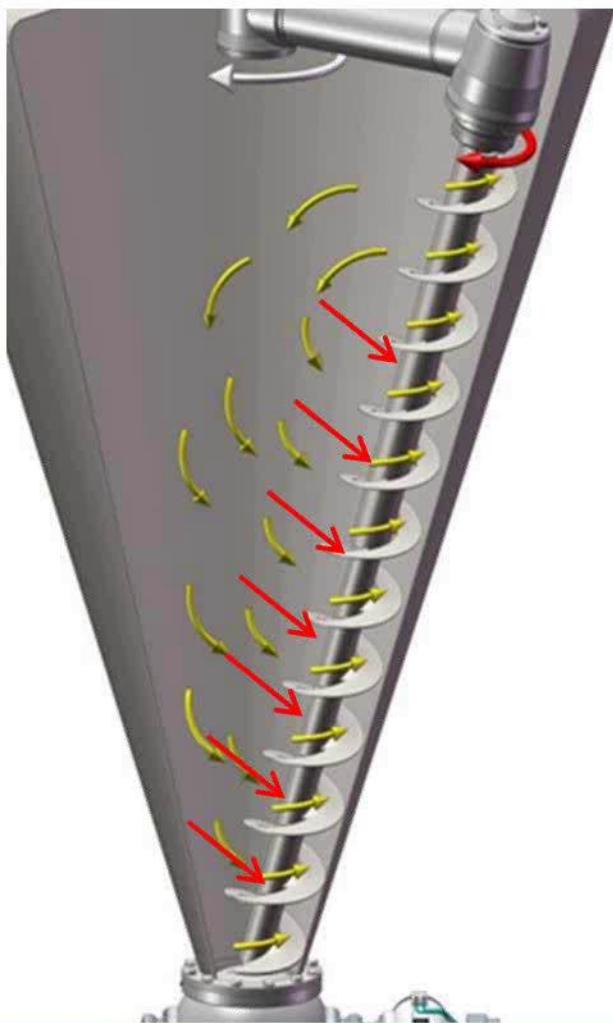


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Convective Transport by Auger



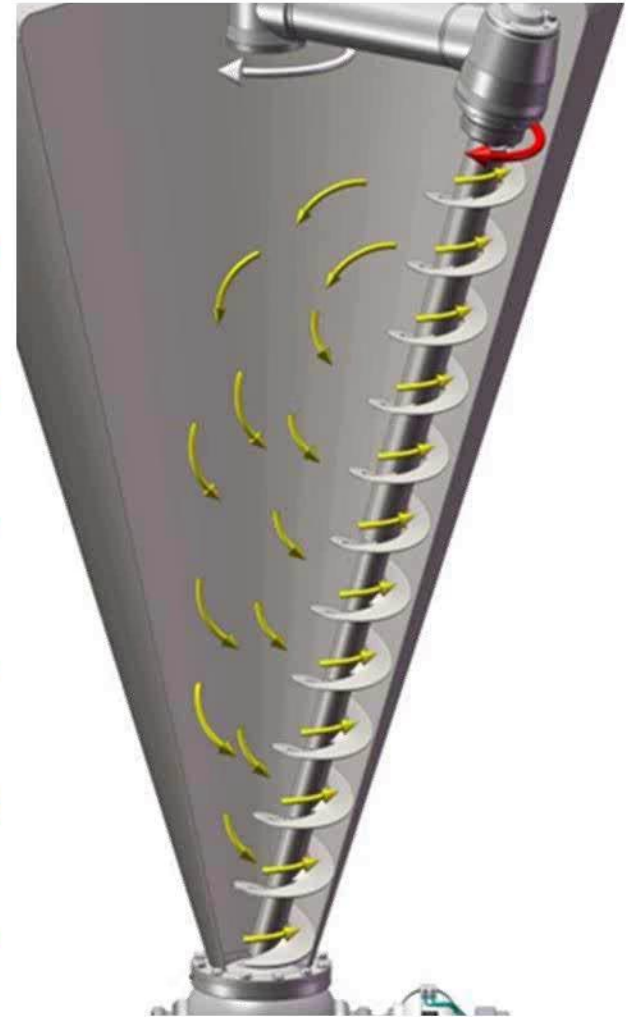
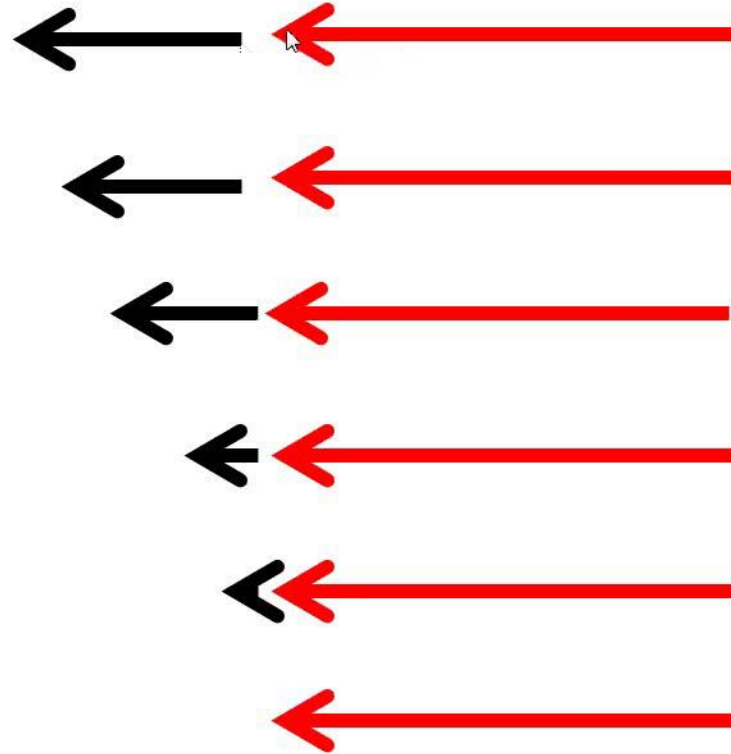
Controlled Randomization



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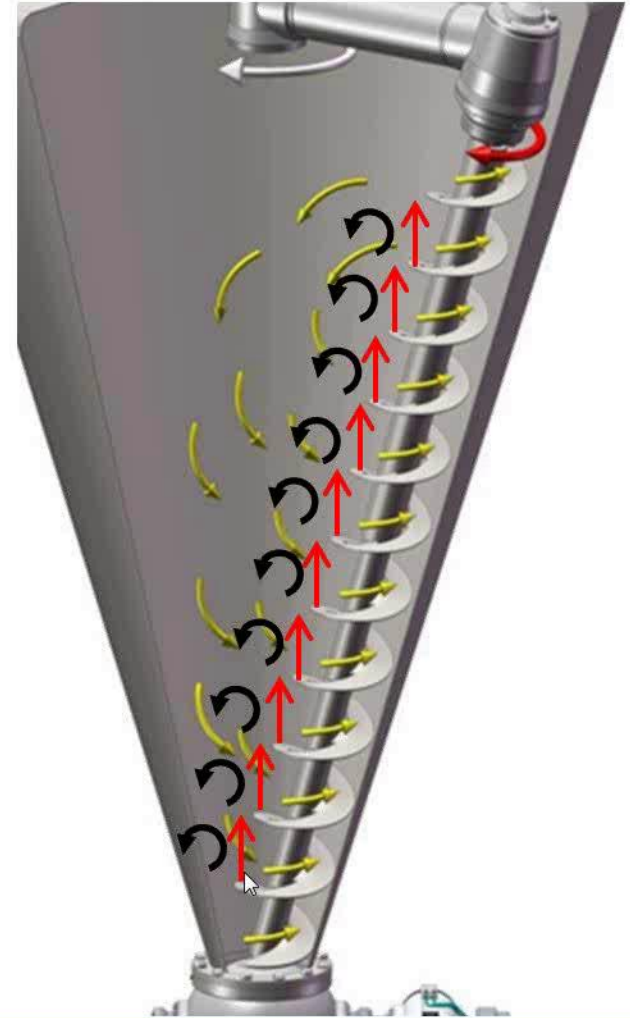
Teamwork = Auger + Orbital Arm

Tip speed
1.0-1.2 m/s
combined
speed

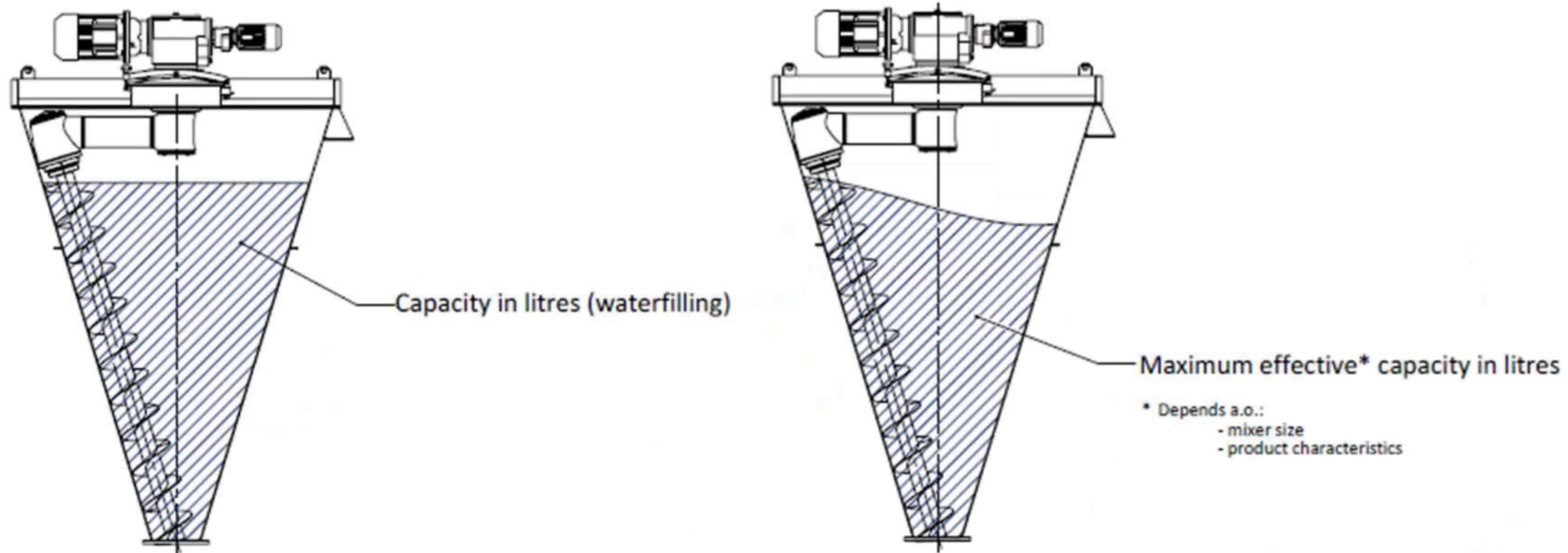


Teamwork = Auger + Orbital Arm

- Auger does more “work” than Orbital arm
- 10:1 ratio of motor sizes



Filling the Mixer



Mixer Capacity is 85-15% of water fill volume



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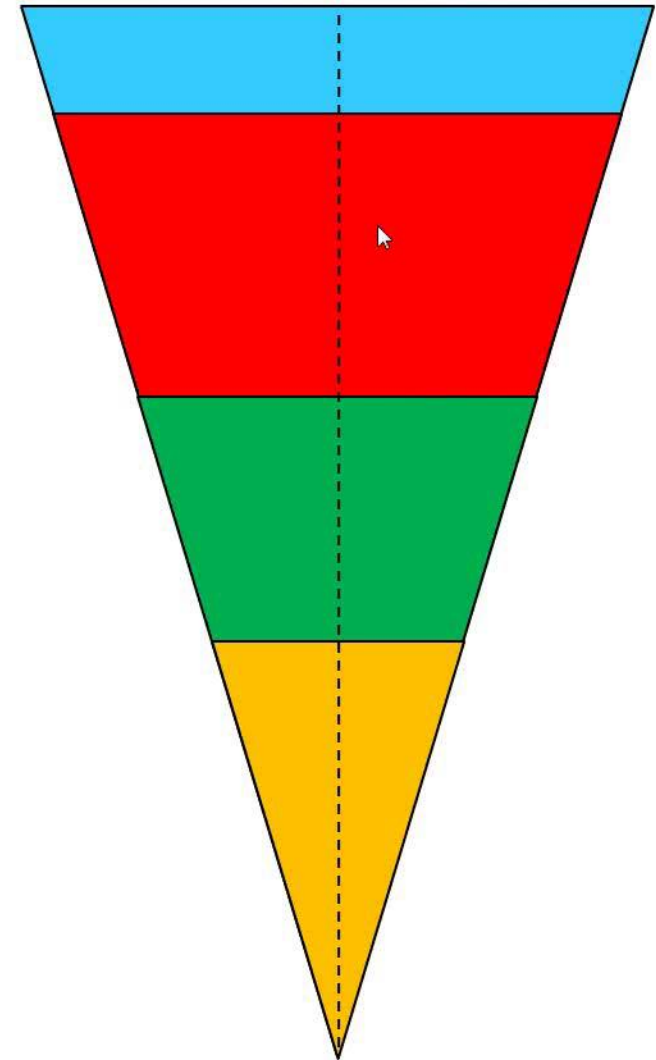
Video of Mixing



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Conical Shape Vessel

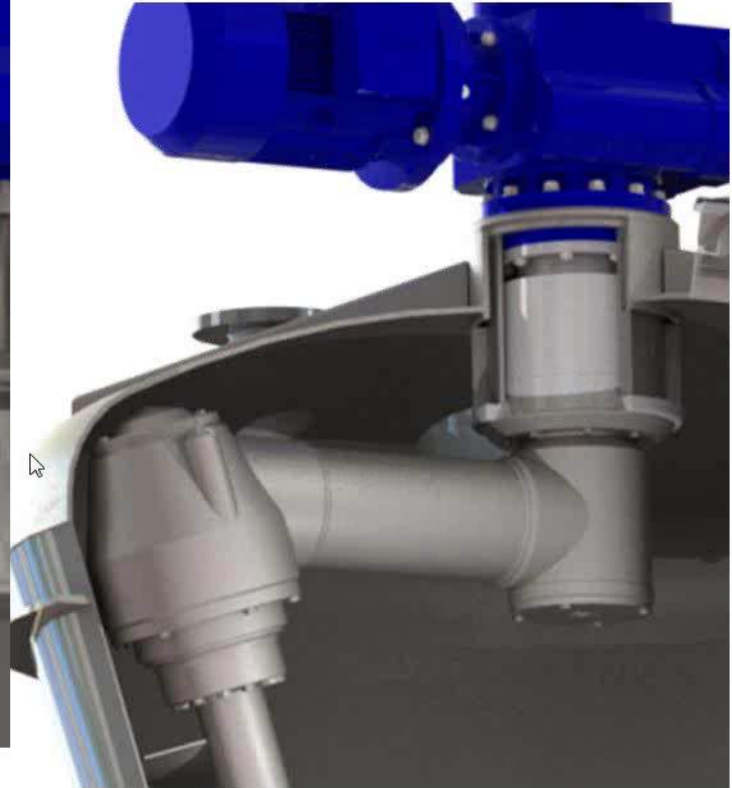
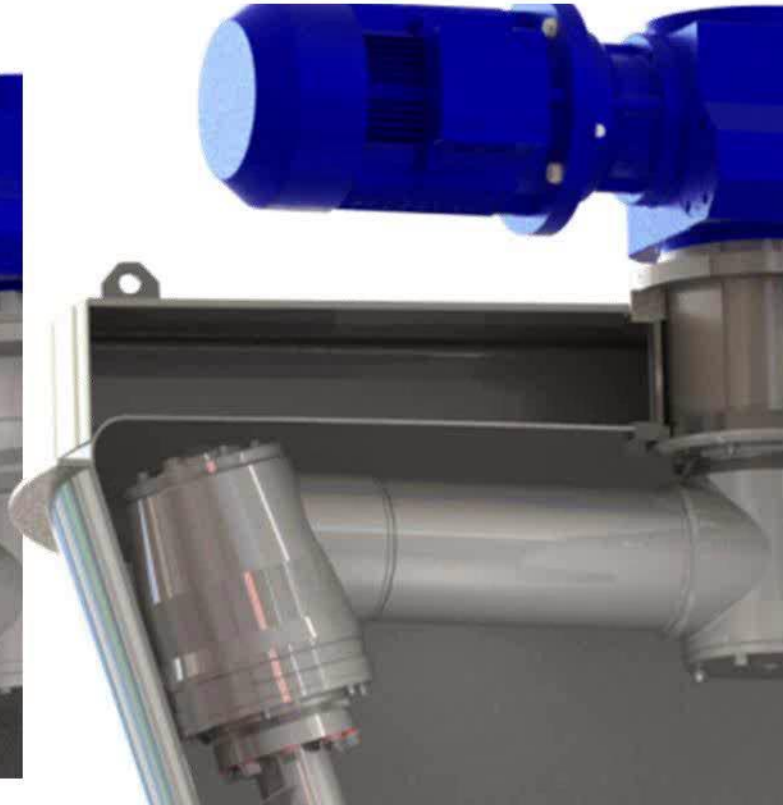
- $V = \pi * r^2 * h / 3$
 - With the bottom angle fixed $r \propto h$
 - $V \propto h^3$, $h \propto V^{1/3}$
 - Double height $\Rightarrow$ 8x more volume
 - In practice it is 8-10x more volume if you double the height



34° (17° from vertical)

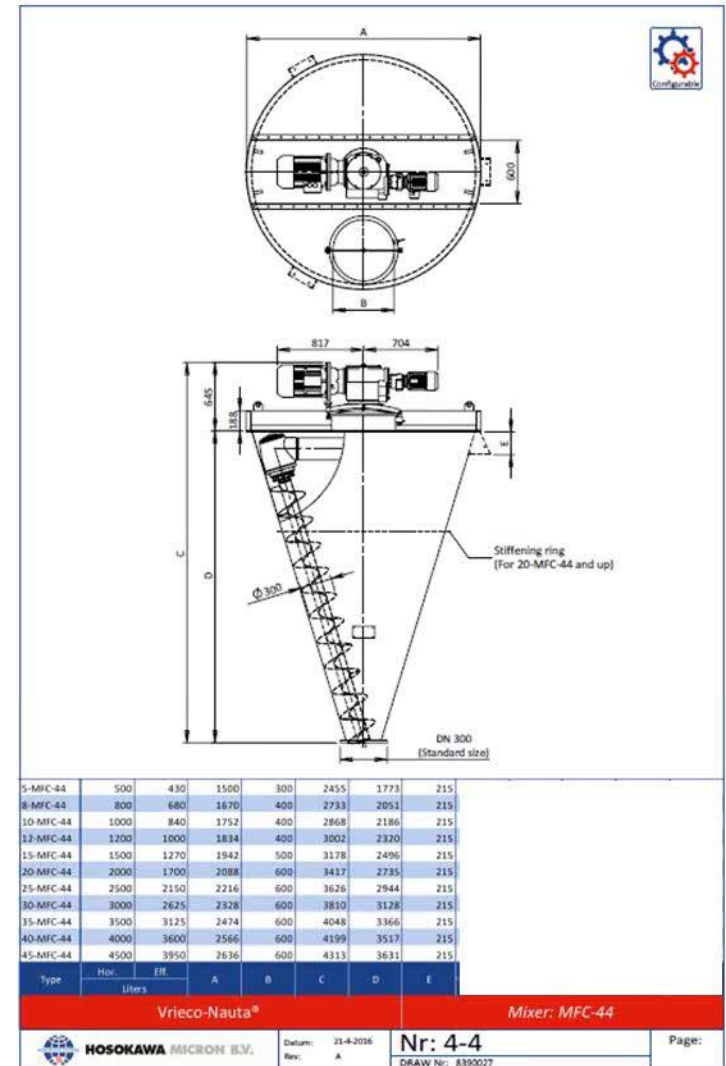


Top Cover



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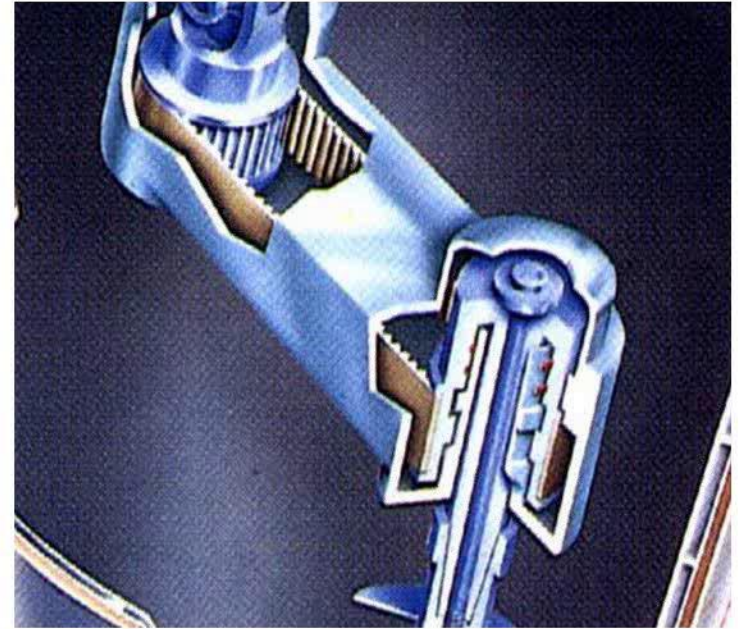
Drives



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Orbital arm

Gear or belt drive
Shape determine type!



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Orbital Arm Sealing Mechanisms

- Different types for different uses
- Two sets of seals
- Types:
 - Lip seal
 - Single Mech.
 - Double Mech.

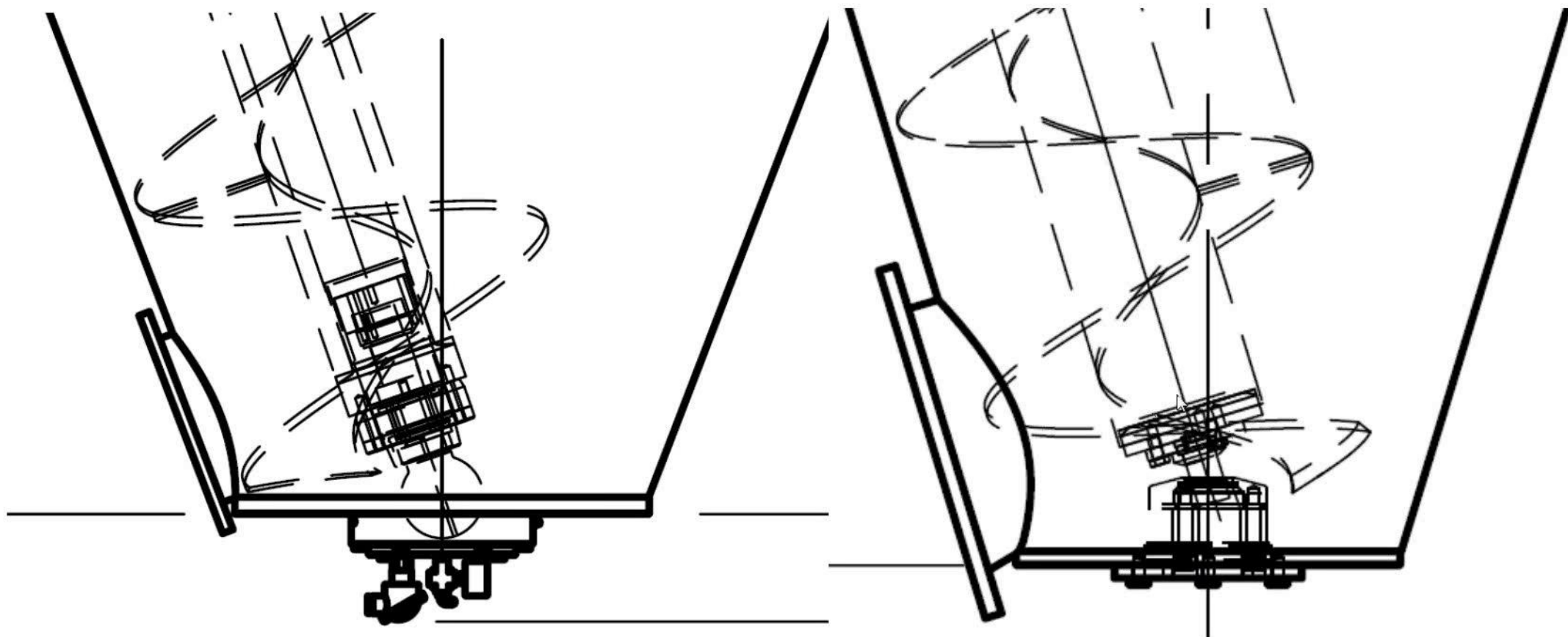


Mixing Screw (Auger)



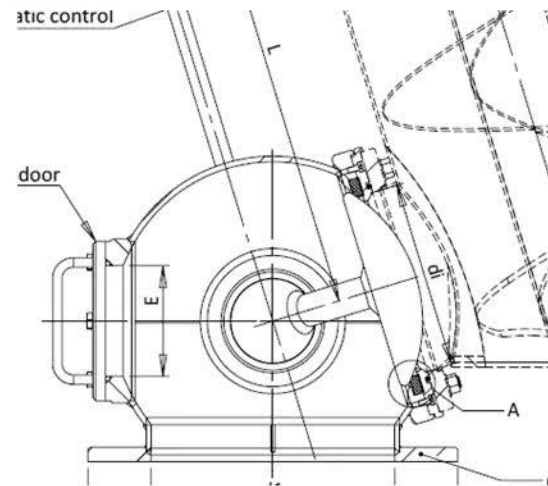
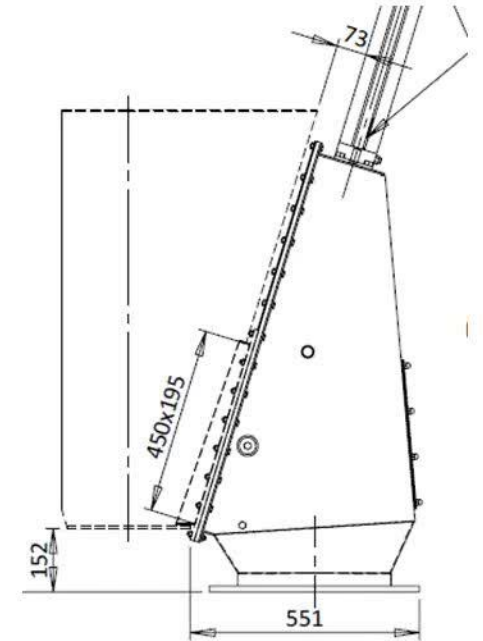
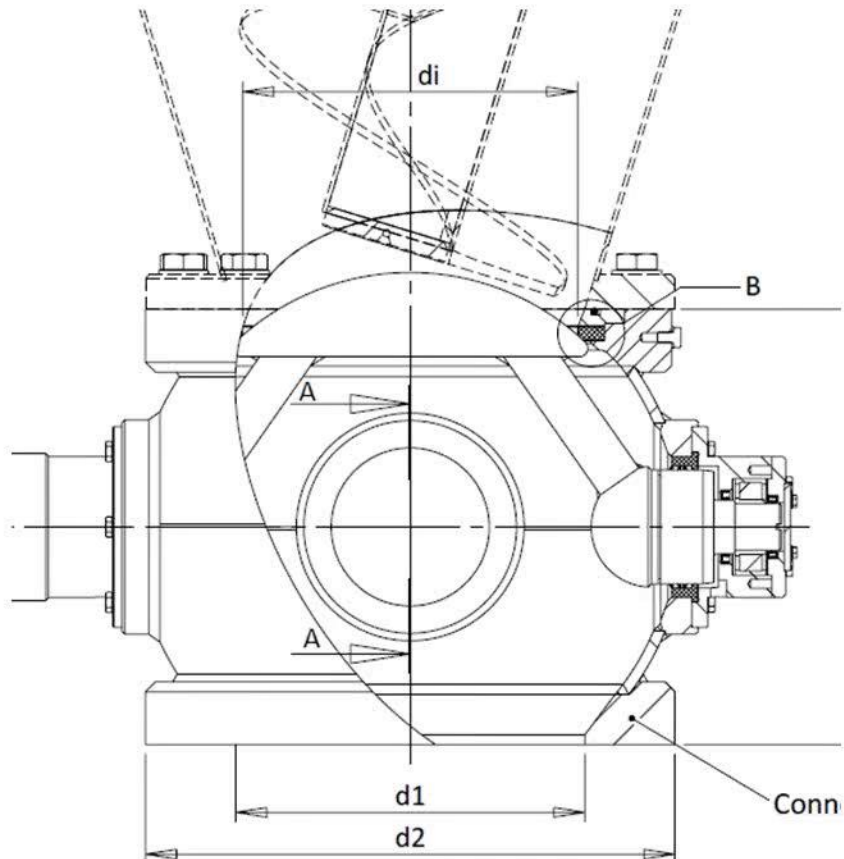
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Supported Mixing Screw



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Discharge Valves

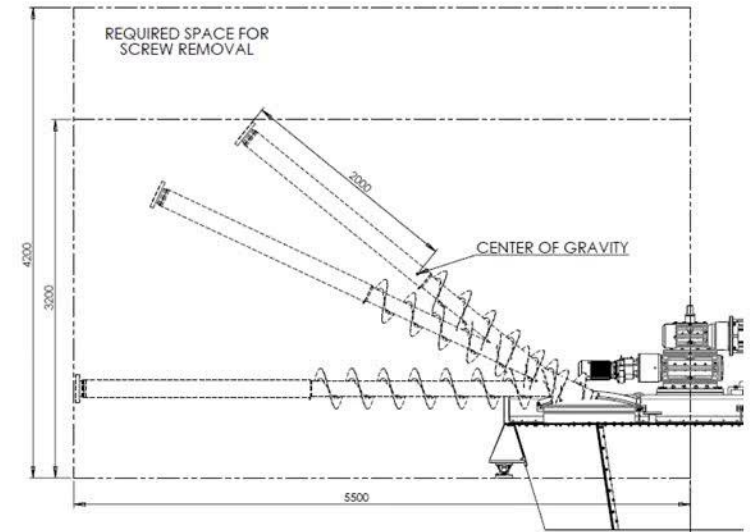
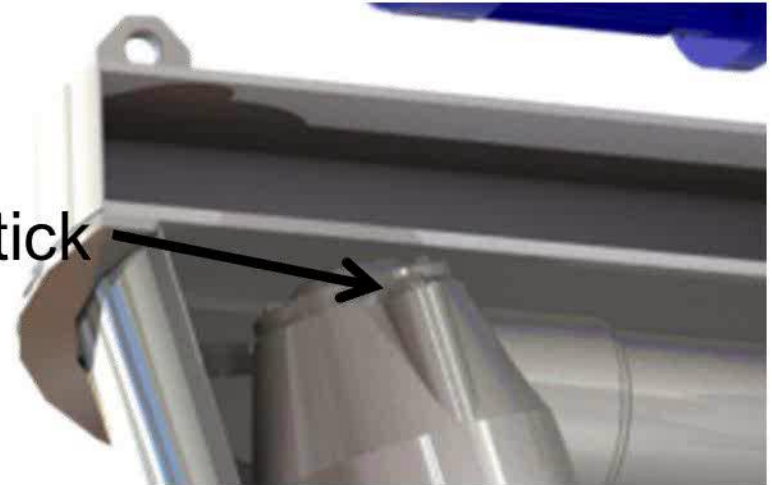


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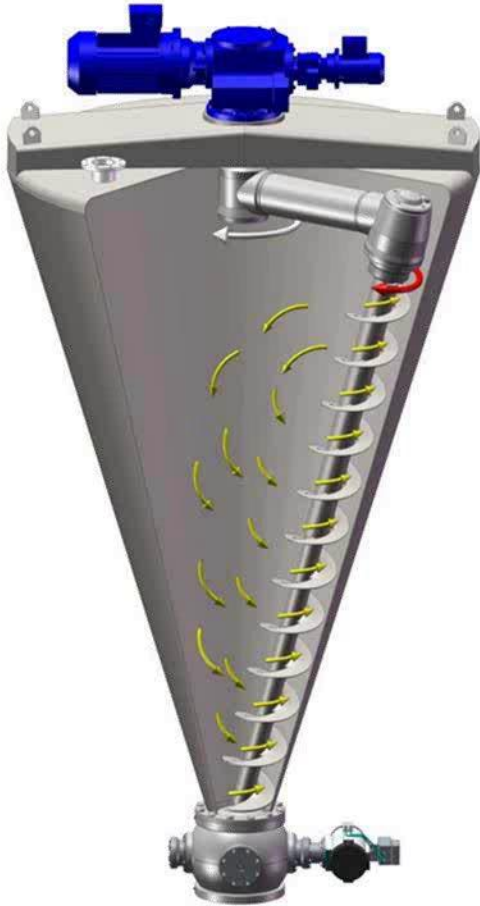
Maintenance

- Check Lubricants
- Change bearings/seals
 - Check height above unit!
 - Removal of top cover sections
 - Removal of entire bridge beam
 - Through inspection hatch
- Service personnel stationed in USA and Europe

Dipstick



Using the Nauta design for drying

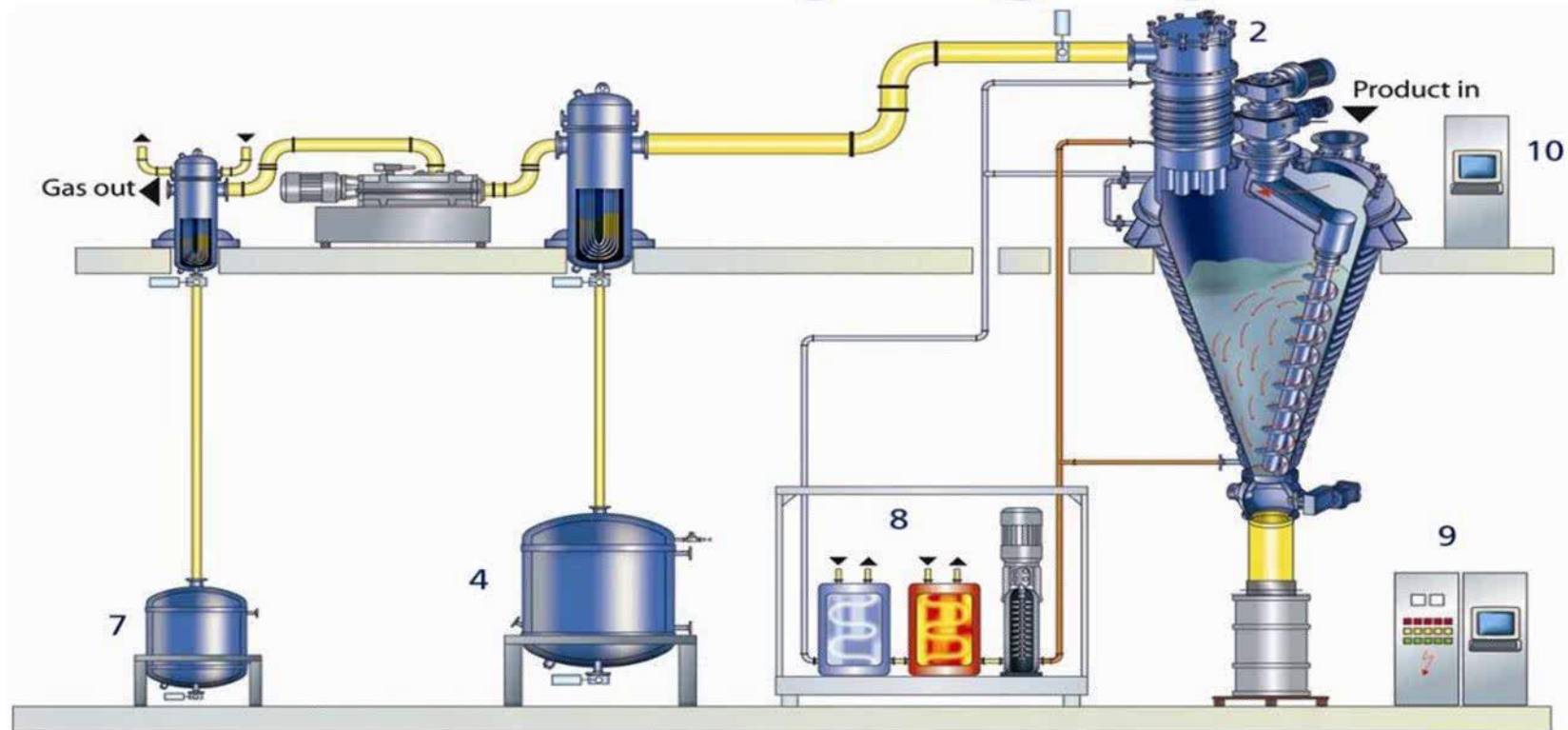


Changes
to



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Vacuum drying system



1. Vacuum dryer
2. Vacuum filter
3. Condenser
4. Receptacle

5. Vacuum pump
6. Optional condenser
7. Optional receptacle
8. Heating cooling skid

9. Control cabinet
10. Operating panel

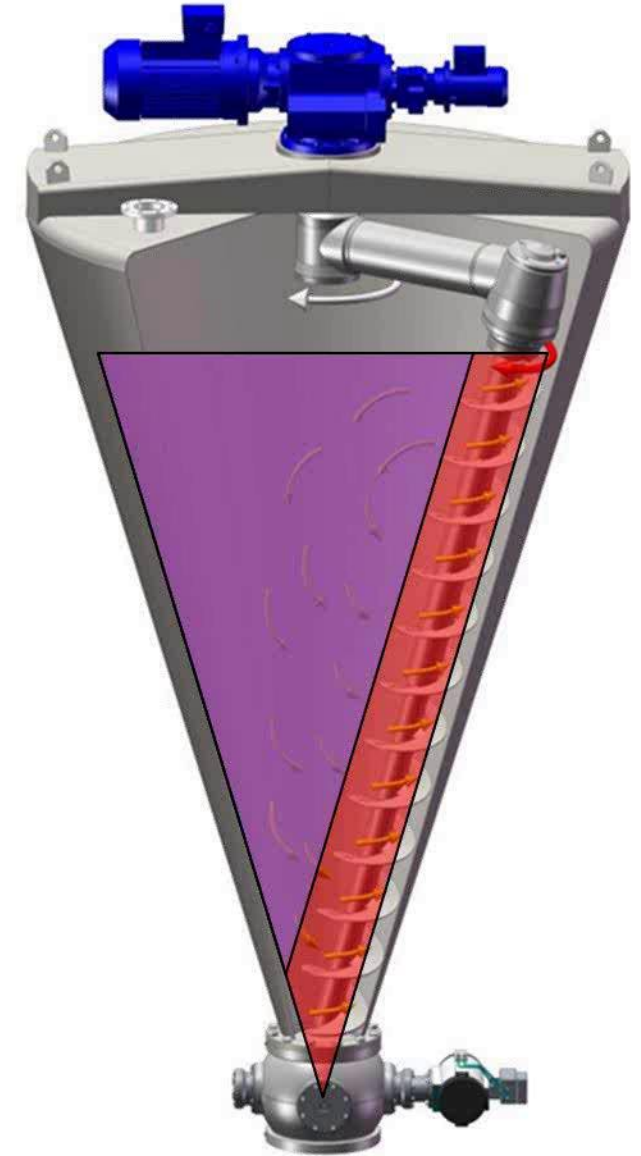


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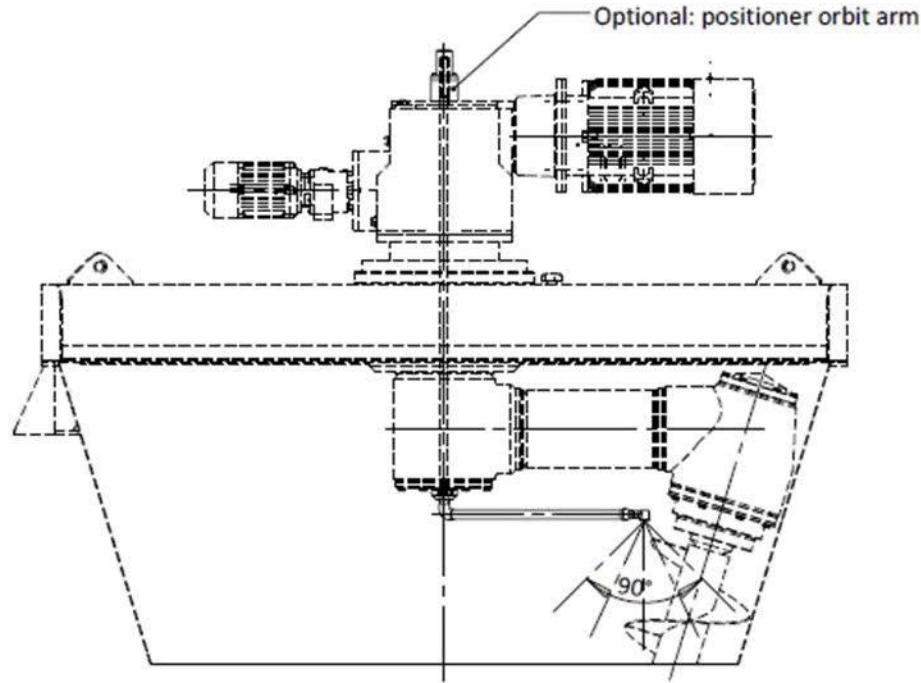
Metal Powders



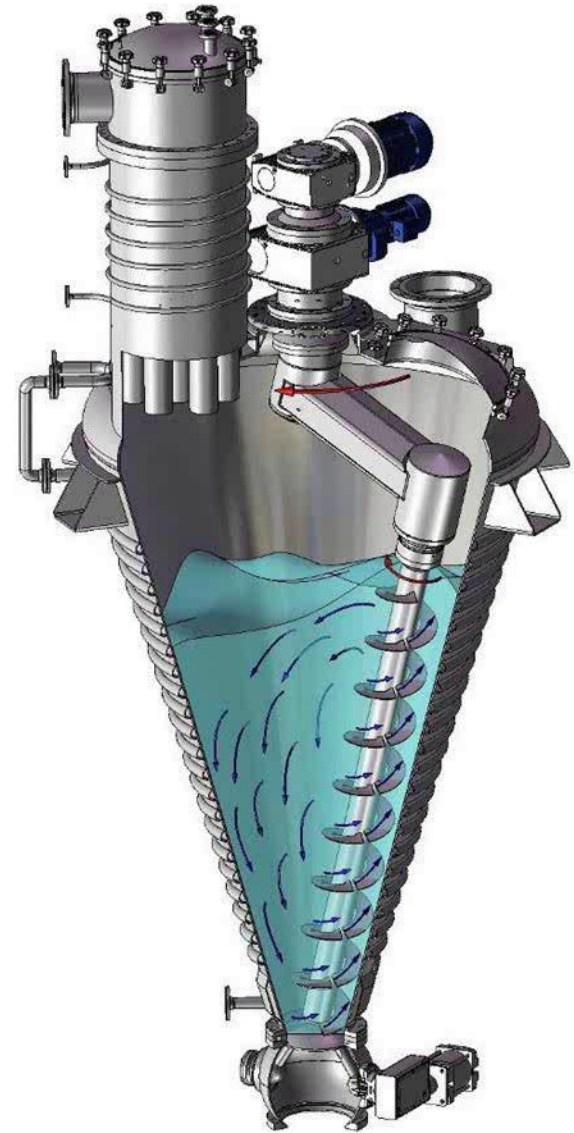
- UO₂ and Fe base powder
- Bulk Density over 2.5 g/mL



Friable Catalyst

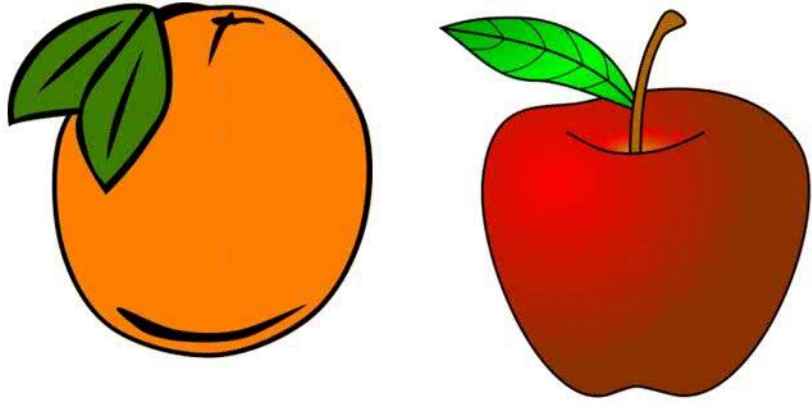


Coating and Drying in a
single vessel

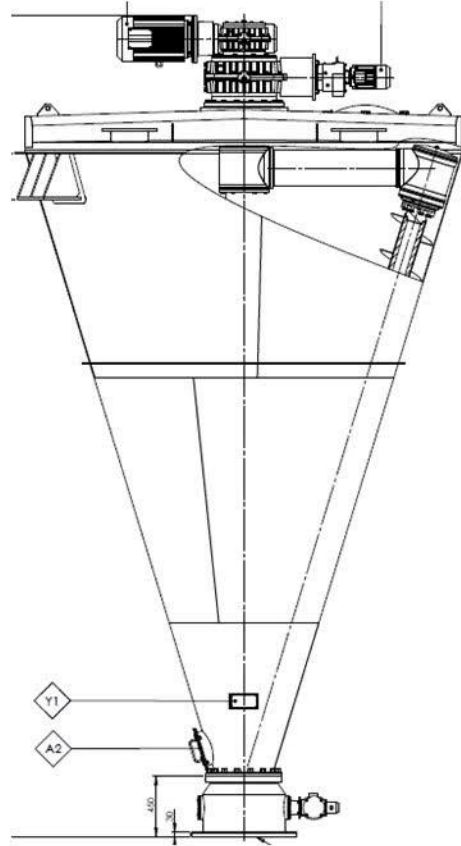


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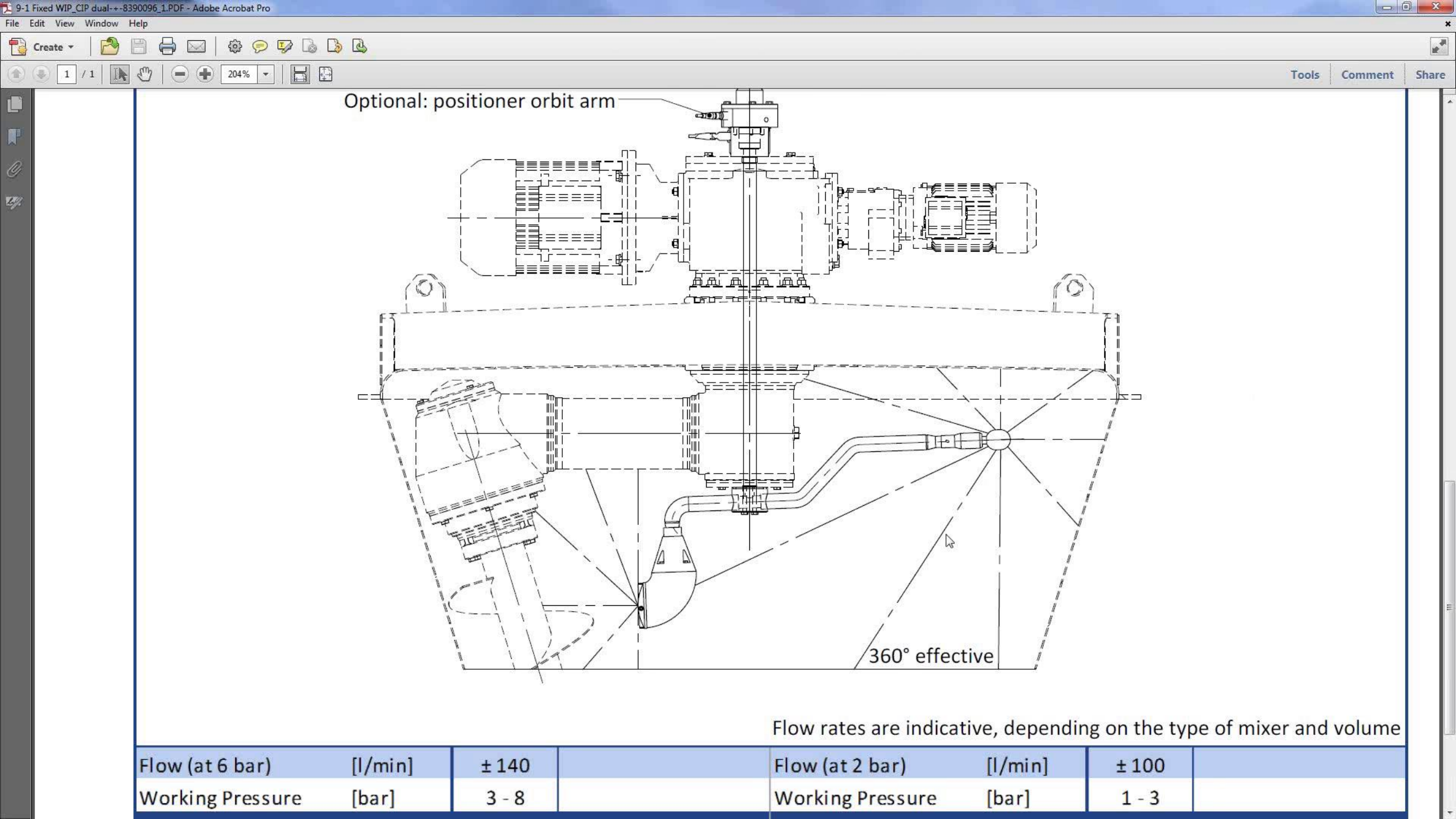
Pectin Blending



- Gentle Blending
- Minimal heat generation even for large batches





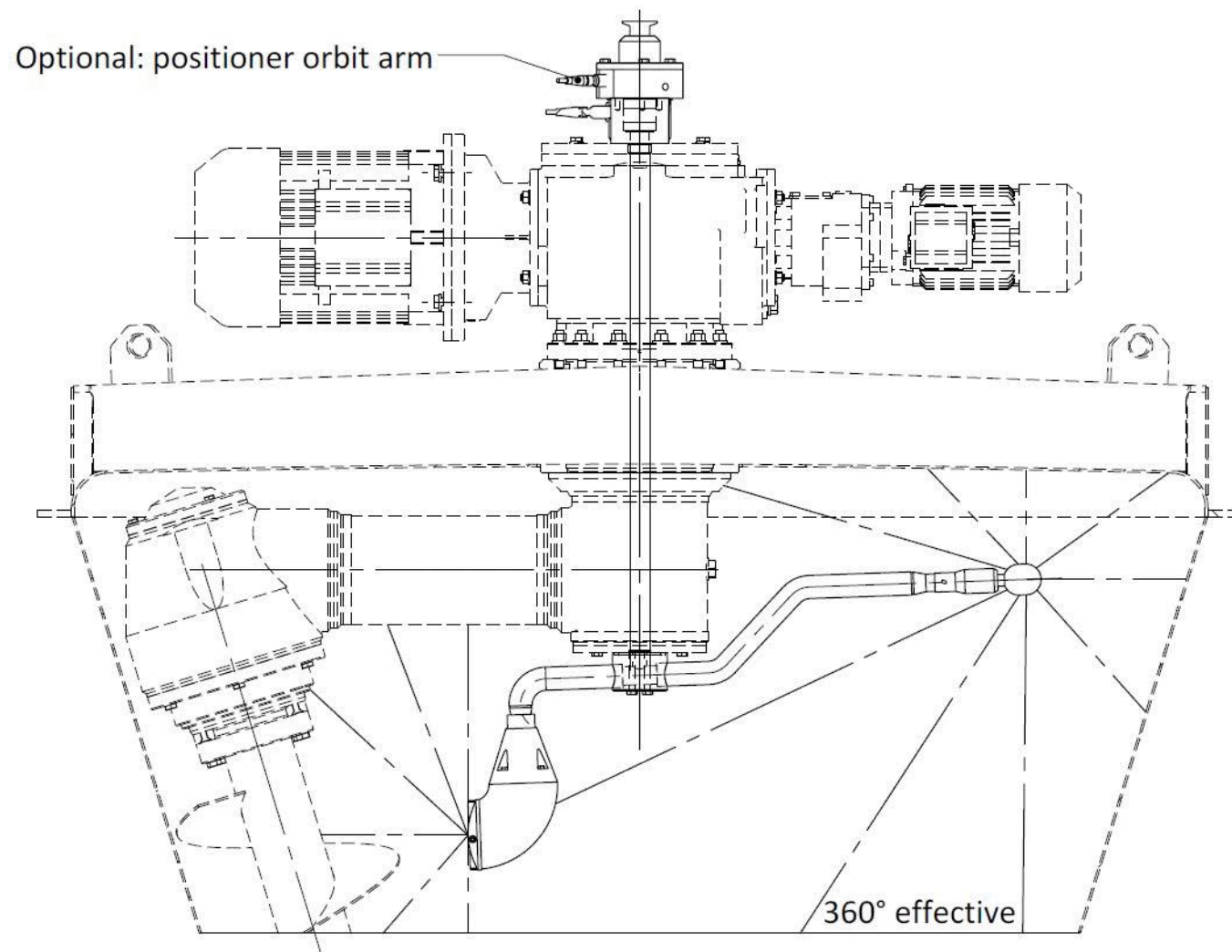


Optional: positioner orbit arm

360° effective

Flow rates are indicative, depending on the type of mixer and volume

Flow (at 6 bar)	[l/min]	± 140	Flow (at 2 bar)	[l/min]	± 100
Working Pressure	[bar]	3 - 8	Working Pressure	[bar]	1 - 3



Flow rates are indicative, depending on the type of mixer and volume

Thank you for your attention!

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